appears in section D (no graphing)

CHAPTER 8

Utility and Demand

ECON101 - Prince Sultan University

OVERVIEW

Learning objectives

The choices a buyer makes are influenced by many factors, which economists summarize as **consumption possibilities** and **preferences**. This chapter uses **marginal utility theory** to explain how a consumer chooses and why demand curves slope downward.

1. Explain the limits to consumption and describe preferences using the concept of utility.
2. Explain the marginal utility theory of consumer choice.
3. Use marginal utility theory to predict the effects of changes in prices and incomes and to explain the paradox of value. → water / diamond
4. Describe some new ways of explaining consumer choices.

OVERVIEW

Running example: Huda's monthly budget

Huda buys only two goods each month: novels and cups of fresh juice.

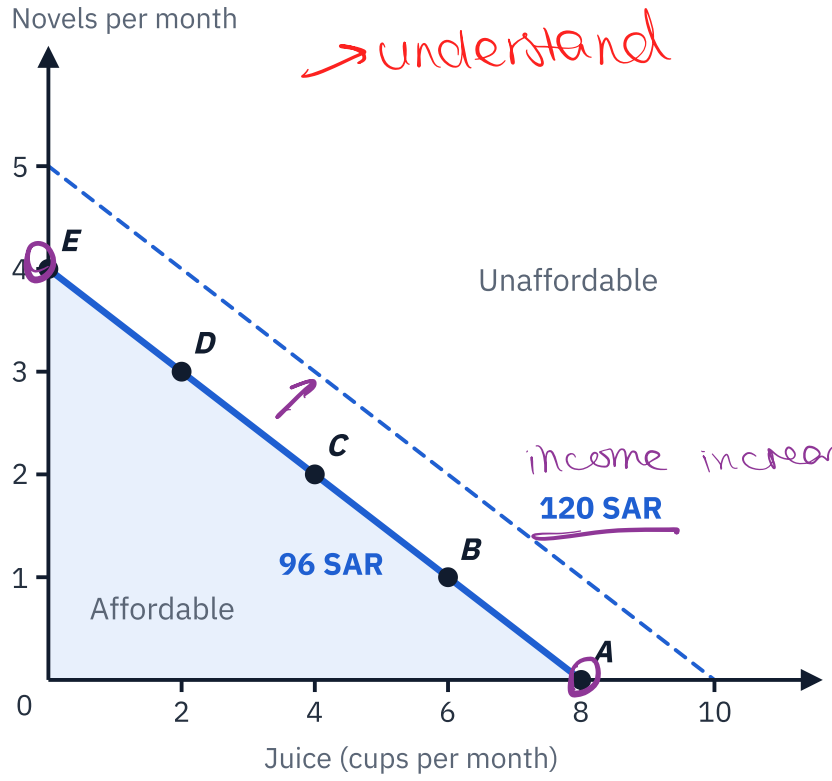
96 SAR
Income to spend each month

24 SAR
Price of a novel (P_N)

12 SAR
Price of a cup of juice (P_J)

Later sections change these prices and her income to predict how her choices respond.

Consumption possibilities



Consumption possibilities are all the things that a consumer can afford to buy. They are limited by **income** and the **prices** of goods.

When Huda spends all her income:

$$24N + 12J = 96$$

Her **budget line** shows the limits of her consumption possibilities.

- Novels and cups of juice are **indivisible goods**, bought in whole units at the points *A* to *E*. A **divisible good** can be bought in any quantity along the line.
- Huda can afford any point **on or inside** her budget line, but not a point outside it.

Changes in consumption possibilities

A rise in income

Shifts the budget line outward and leaves its slope unchanged. With 120 SAR, Huda can buy up to 5 novels or 10 cups (the dashed line).

A rise in a price (rotates)

Changes the slope of the line and shrinks consumption possibilities. If juice rises to 24 SAR, Huda can afford at most 4 cups instead of 8.

The budget line shows what is possible; preferences determine which possibility is chosen.

- budget → possibilities
- preferences → determining which possibilities are chosen

Preferences and utility

Huda's choice depends on her **preferences**, her likes and dislikes. The benefit or satisfaction from consuming a good or service is called **utility**.

Total utility

The total benefit a person gets from the consumption of goods. Generally, more consumption gives more total utility.

Marginal utility

The change in total utility that results from a one-unit increase in the quantity of a good consumed.

Diminishing marginal utility

As the quantity consumed of a good increases, the marginal utility from it decreases.

CONSUMPTION CHOICES

Huda's utility schedules

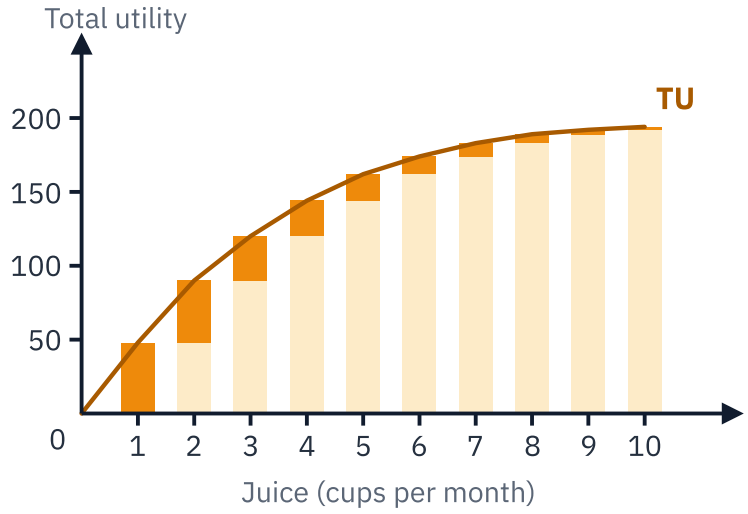
Quantity per month	Novels: total utility	Novels: marginal utility	Juice: total utility	Juice: marginal utility
0	0	–	0	–
1	60	60	48	48
2	108	48	90	42
3	150	42	120	30
4	186	36	144	24
5	216	30	162	18
6	240	24	174	12
7	258	18	183	9
8	270	12	189	6
9	279	9	192	3
10	285	6	194	2

Units of utility per month.

For example, the 3rd cup of juice raises total utility from 90 to 120 units, so its marginal utility is 120 – 90 = **30 units**.

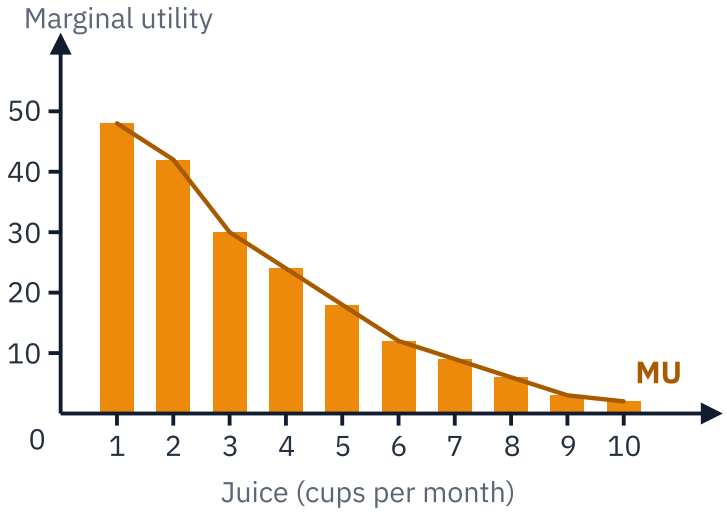
Each marginal utility is smaller than the one before it: **diminishing marginal utility**.

Graphing total and marginal utility from juice



(a) Total utility

Total utility from juice increases as Huda drinks more. The dark top of each bar is the marginal utility of that cup.



(b) Marginal utility

The same marginal utilities get smaller as she drinks more: **diminishing marginal utility**.

A spreadsheet solution

The key assumption is that the consumer chooses the consumption possibility that **maximizes total utility**.

1. Find the **just-affordable combinations**: those that spend all 96 SAR ($24N + 12J = 96$).
2. Find the **total utility** for each combination.
3. The combination with the highest total utility is the consumer's choice.

Row	Novels	Utility from novels	Juice (cups)	Utility from juice	Total utility
A	0	0	8	189	189
B	1	60	6	174	234
C	2	108	4	144	**252**
D	3	150	2	90	240
E	4	186	0	0	186

UTILITY-MAXIMIZING CHOICE

Consumer equilibrium

Consumer equilibrium is a situation in which a consumer has allocated all of the available income in the way that maximizes total utility, given the prices of goods and services.

2	4	252
novels a month	cups of juice a month	units of total utility: 108 + 144

Huda maximizes total utility in row *C* of the spreadsheet.

Choosing at the margin

A more natural way to find the consumer equilibrium is to make choices **at the margin**: having made a choice, would spending a riyal more or a riyal less on a good bring more total utility?

Marginal utility per riyal

The marginal utility from a good that results from spending one more riyal on it. It equals the good's marginal utility divided by its price: MU_N/P_N for novels and MU_J/P_J for juice.

Comparing MU_N/P_N with MU_J/P_J shows whether the budget is allocated in the way that maximizes total utility.

UTILITY-MAXIMIZING CHOICE

The utility-maximizing rule

A consumer's total utility is maximized by following the rule:

1

Spend all available income.

2

Equalize the marginal utility per riyal for all goods.

UTILITY-MAXIMIZING CHOICE

Huda's marginal calculation

3

$P_N = 24$

$P_J = 12$

Each row is a just-affordable combination. *MU* is the marginal utility of the last unit bought.

Row	Novels	MU_N	MU_N/P_N	Juice (cups)	MU_J	MU_J/P_J
A	0	–	–	8	6	6/12 0.50
B	1	60	60/24 2.50	6	12	12/12 1.00
C	**2**	**48**	48/24 **2.00**	**4**	**24**	24/12 **2.00**
D	3	42	42/24 1.75	2	42	42/12 3.50
E	4	36	36/24 1.50	0	–	0/12 = 0 –

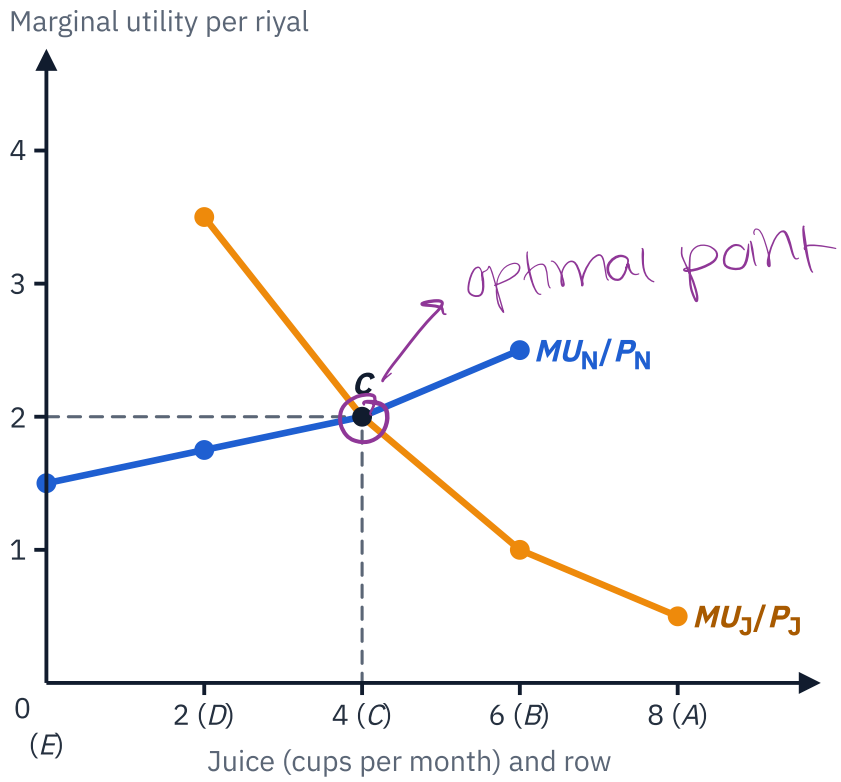
In row *C*, $MU_N/P_N = 48/24 = 2$ and $MU_J/P_J = 24/12 = 2$: Huda spends all her income and the marginal utility per riyal is equal for both goods.

Utility Maximization

$$\frac{MU_N}{P_N} = \frac{MU_J}{P_J}$$

$24(2) + 12(4) = 96$

Why the rule works



Marginal utility per riyal along the budget line. Moving right means more juice and fewer novels.

Row *B*: $MU_J/P_J = 1 < MU_N/P_N = 2.5$. Too much on juice, too little on novels. Spending less on juice and more on novels increases MU_J and decreases MU_N.

Row *D*: $MU_J/P_J = 3.5 > MU_N/P_N = 1.75$. Too little on juice, too much on novels. Spending more on juice and less on novels decreases MU_J and increases MU_N.

Row *C*: $MU_J/P_J = MU_N/P_N = 2$. Huda is maximizing utility.

UTILITY-MAXIMIZING CHOICE

Checking with total utility

+18

B → ***C*****:** Huda gives up the 6th and 5th cups (12 + 18 = 30 units) and gains the 2nd novel (48 units). Total utility rises by 18.

−12

C → ***D*****:** she gives up the 4th and 3rd cups (24 + 30 = 54 units) and gains the 3rd novel (42 units). Total utility falls by 12.

PREDICTIONS

A fall in the price of a novel

Marginal utility theory predicts how choices respond to changes in prices and income.

The price of a novel falls from 24 SAR to 12 SAR (juice stays at 12 SAR; income stays at 96 SAR). At Huda's old choice of 2 novels and 4 cups, 24 SAR is left unspent and:

4	**>**	**2**
$MU_N/P_N = 48/12$		$MU_J/P_J = 24/12$

To restore equality, she buys more novels, which lowers MU_N, and less juice, which raises MU_J.

PREDICTIONS

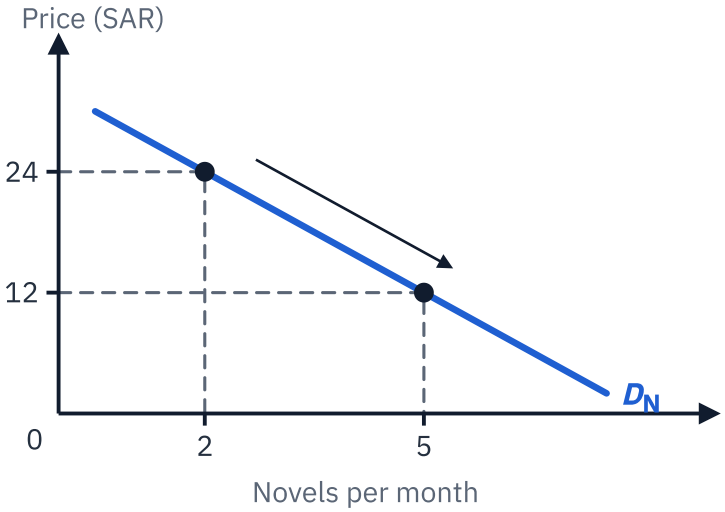
The new choice after the price of a novel falls

Novels	Utility	MU_N/P_N	Juice (cups)	Utility	MU_J/P_J	Total utility
0	0	–	8	189	0.50	189
1	60	5.00	7	183	0.75	243
2	108	4.00	6	174	1.00	282
3	150	3.50	5	162	1.50	312
4	186	3.00	4	144	2.00	330
5	216	2.50	3	120	2.50	**336**
6	240	2.00	2	90	3.50	330
7	258	1.50	1	48	4.00	306
8	270	1.00	0	0	–	270

New consumer equilibrium: 5 novels and 3 cups of juice, where $MU_N/P_N = MU_J/P_J = 2.5$ and total utility is 336 units.

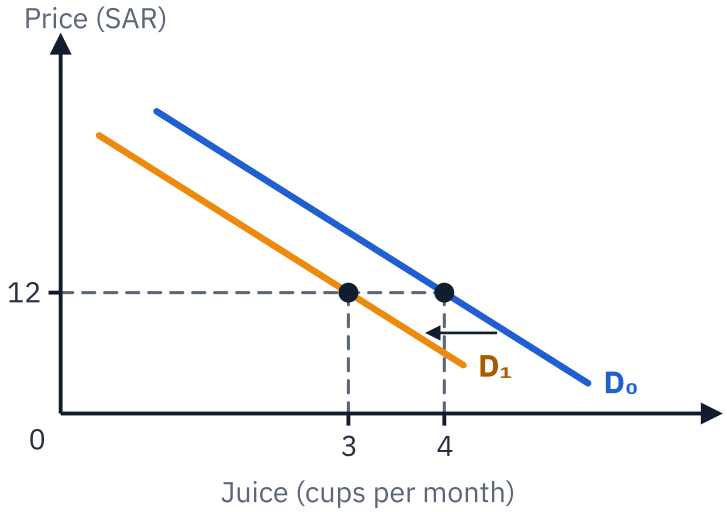
PREDICTIONS

A fall in the price of a novel: demand curves



(a) Demand for novels

The quantity of novels demanded rises from 2 to 5: a **movement along** the demand curve, which slopes downward.



(b) Demand for juice

Demand for juice **decreases**: at 12 SAR, Huda buys 3 cups instead of 4. The curve shifts leftward from D_0 (novels at 24 SAR) to D_1 (novels at 12 SAR).

For Huda, novels and juice are **substitutes**. Only the dots are calculated; the lines are illustrative.

PREDICTIONS

A rise in the price of juice



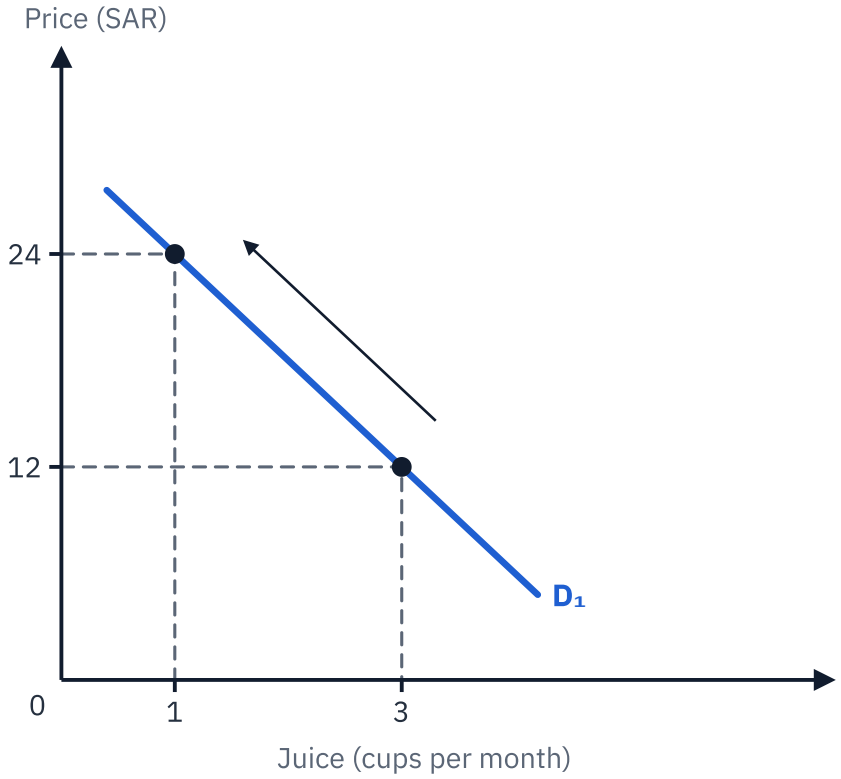
The price of juice rises from 12 SAR to 24 SAR (a novel costs 12 SAR; income is 96 SAR). Her previous choice of 5 novels and 3 cups now costs 132 SAR, more than her income. At that choice, $MU_J/P_J = 30/24 = 1.25 < MU_N/P_N = 30/12 = 2.5$, so Huda **drinks less juice**, which raises MU_J, and **buys more novels**, which lowers MU_N.

Novels	Utility	MU_N/P_N	Juice (cups)	Utility	MU_J/P_J	Total utility
8	270	1.00	0	0	–	270
6	240	2.00	1	48	2.00	**288**
4	186	3.00	2	90	1.75	276
2	108	4.00	3	120	1.25	228
0	0	–	4	144	1.00	144

New consumer equilibrium: 6 novels and 1 cup of juice, where $MU_N/P_N = 24/12 = 2$ and $MU_J/P_J = 48/24 = 2$.

PREDICTIONS

A rise in the price of juice: demand



- A rise in the price of juice **decreases the quantity of juice demanded** from 3 cups to 1: a **movement along** the demand curve for juice.
- The demand for novels **increases**: at 12 SAR a novel, Huda now buys 6 instead of 5, so the demand curve for novels shifts rightward.

Only the dots are calculated; the line through them is illustrative.

PREDICTIONS

A rise in income

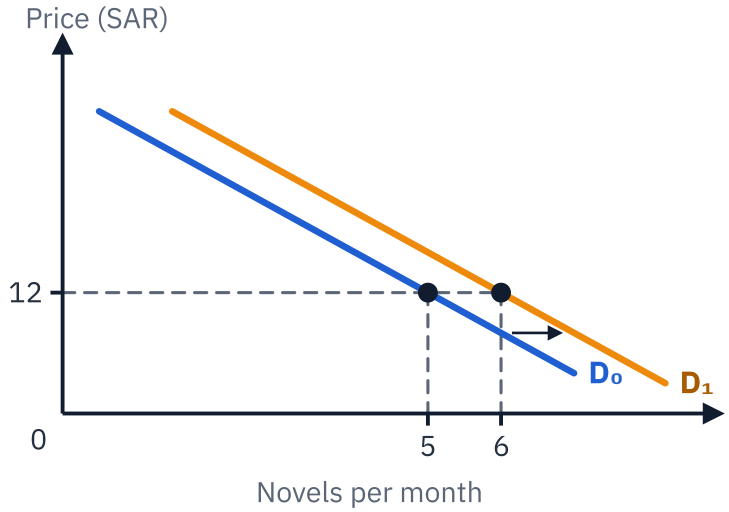
When income increases, the **demand for a normal good increases**. With both prices at 12 SAR, Huda's income rises from 96 SAR to 120 SAR. To spend all her income, she uses the extra 24 SAR. The just-affordable combinations near the best one are:

Novels	Utility	MU_N/P_N	Juice (cups)	Utility	MU_J/P_J	Total utility
4	186	3.00	6	174	1.00	360
5	216	2.50	5	162	1.50	378
6	240	2.00	4	144	2.00	**384**
7	258	1.50	3	120	2.50	378
8	270	1.00	2	90	3.50	360

Huda now buys 6 novels and 4 cups instead of 5 and 3. She buys more of both goods, so for her novels and juice are normal goods.

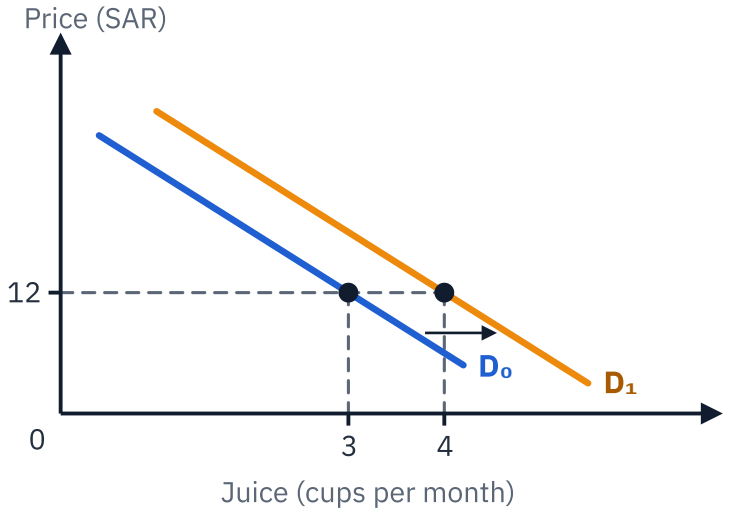
PREDICTIONS

A rise in income: both demand curves shift rightward



(a) Novels

At 12 SAR, Huda buys 6 novels instead of 5.



(b) Juice

At 12 SAR, Huda buys 4 cups instead of 3.

D_0 is for income 96 SAR and D_1 for 120 SAR; prices stay at 12 SAR. Only the dots are calculated; the lines are illustrative.

PREDICTIONS

Summary of the predictions

Situation	P_N (SAR)	P_J (SAR)	Income (SAR)	Novels	Juice (cups)
Starting point	24	12	96	2	4
Price of a novel falls	12	12	96	5	3
Price of juice rises	12	24	96	6	1
Income rises	12	12	120	6	4

The last two rows start from the second row (both prices at 12 SAR, income 96 SAR). In each case, Huda spends all her income and equalizes the marginal utility per riyal. Marginal utility theory predicts:

- When the price of a good falls, the quantity demanded of that good increases: the demand curve slopes downward.
- A change in the price of one good changes the demand for another good.
- When income increases, the demand for a normal good increases.

PREDICTIONS

Temperature: an analogy

Utility is similar to temperature. Both are abstract concepts, and both have units of measurement that are arbitrary.

The concept of utility helps us to predict consumption choices in the same way that the concept of temperature enables us to predict when water will turn to ice or steam. It helps us understand why:

People buy more of a good when its price falls.

People buy more of most goods (normal) when their income increases.

PREDICTIONS

The paradox of value

Water, which is essential to life, is far cheaper than diamonds, which are not essential.

The paradox is resolved by distinguishing between **total utility** and **marginal utility**.

Water

We use so much water that the marginal utility from the water we consume is small, but its total utility is large.

Diamonds

We buy few diamonds, so the marginal utility from diamonds is large, but their total utility is small.

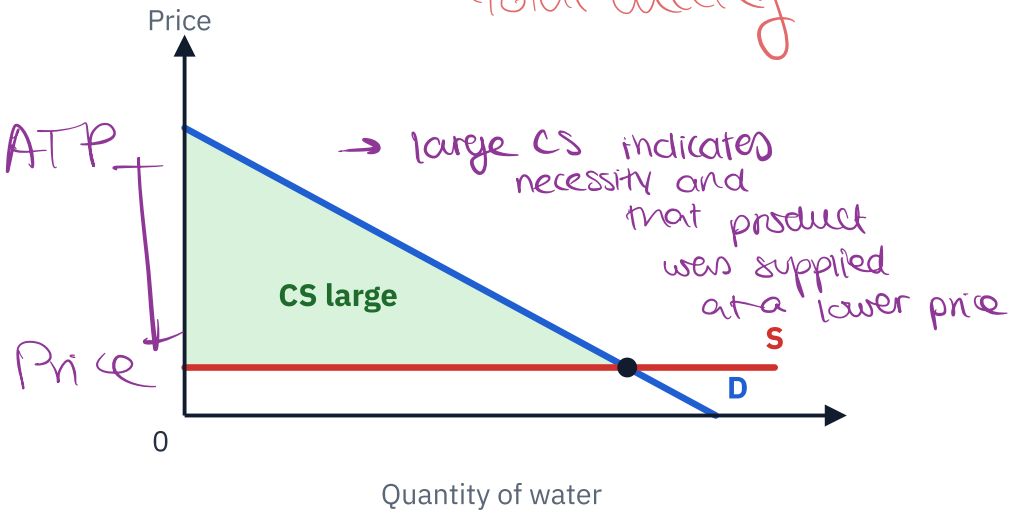
Consumers maximize utility, so the marginal utility per riyal is equal for water and diamonds. A high price goes with a high marginal utility.

PREDICTIONS

Value and consumer surplus

- MCQs / theory

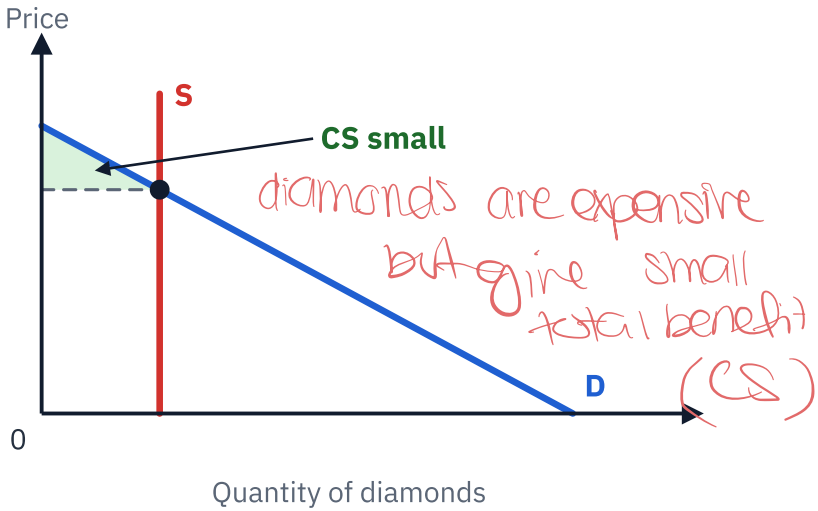
total utility



→ large CS indicates necessity and that product was supplied at a lower price

(a) Water

Supply is **perfectly elastic** at a low price, so the quantity consumed is large and the consumer surplus from water is large.



diamonds are expensive but give small total benefit (CS)

(b) Diamonds →

Supply is **perfectly inelastic** at a small quantity, so the price is high and the consumer surplus from diamonds is small.

water is cheap but gives greater total benefit (CS)

NEW WAYS OF EXPLAINING CHOICES

Behavioral economics

Behavioral economics studies the ways in which limits on the human brain's ability to compute and implement rational decisions influence economic behavior: both the decisions that people make and the consequences of those decisions for the way markets work. There are three impediments to rational choice:

Bounded rationality

Rationality that is bounded by the computing power of the human brain. Faced with uncertainty, people rely on rules of thumb, the views of others, or gut instinct.

Bounded willpower

Less-than-perfect willpower: it does not stop us from making a decision that we know, at the time of implementing it, we will later regret.

Bounded self-interest

The limited self-interest that sometimes results in suppressing our own interests to help others.

The main applications are in **finance**, where uncertainty is the key factor, and **savings**, where the future is the key factor.

The endowment effect and neuroeconomics

The endowment effect

The tendency for people to value something more highly simply because they own it. A student who would pay at most 150 SAR for a pair of headphones may refuse to sell the same pair for 200 SAR once she owns it.

Neuroeconomics is the study of the activity of the human brain when a person makes an economic decision. Different decisions appear to activate different areas of the brain:

Prefrontal cortex

Where memories are stored and data analyzed.
Decisions made here might be deemed **rational**.

Hippocampus

Where memories of anxiety and fear are stored.
Decisions made here might be deemed **irrational**.

The controversy

Should economics focus on explaining the decisions we observe, or on what goes on inside people's heads?

For most economists, the goal of economics is to **explain the decisions that we observe people make**, not to explain what goes on inside people's heads.

Source

Michael Parkin (2022) *Microeconomics, Global Edition*, 14th Edition, published by Pearson Education.